NUS School of Computing

FT5007 Capstone

Month < June 2026 > Report by Individual Team Member

Project Title: Agentic AI Chatbot for Consumer Banking

Reporting Period: June 2026

For each team member, please describe your achievements, upcoming plans (to-dos), challenges, and plans to overcome challenges in this period.

Student Name: Wu Qianzong

Achievement during the month:

- **Designed Macro Product Scope:** Designed a three-portal architecture (Customer-Facing Mobile Portal, Human Agent Escalation Portal, and Backend Admin Config Panel) for the agentic AI chatbot.
- **Defined MVP Boundaries:** Narrowed the scope strictly to the Customer-Facing Mobile Portal for MVP. This included narrowing down to 3 intent families (Account Opening, Digital Banking, and Profile), enforcing authenticated users only for MVP.
- **Finalized Scope Documents:** Completed (to version 3.5) the Product scope, Data scope (FAQ data preparation, sources and synthetic data generation method), Technical Architecture(Local DB, State Mng, web+Remote LLM), and Evaluation Metrics for the MVP.
- **Technical Feasibility Check:** Validated the technologies used in the MPV to ensure the architecture runs successfully on a local 16GB M1 Mac by offloading generative tasks to a remote LLM (Groq API) while keeping the RAG vector database (ChromaDB), state management and webserver local.
- **MVP UI Demo:** Built a front-end React MVP demo showcasing progressive disclosure, precision substitution, deep-linking, and the human escalation with case summary.

To-dos in next month:

The whole objective is to achieve end-to-end basic conversational capability for one intent (account opening) next month and build and verify the basic infrastructure needed to support the whole process.

Key Implementation Tasks:

1. Data & User

- **Build Knowledge Base of FAQ:** Construct the local RAG pipeline by ingesting the FAQ documents to allow the chatbot to retrieve general policy details. This includes building automatic retrieval functions to retrieve publicly available FAQ data in the bank website.
- **Implement Mock User Switch:** Set up the predefined profile-switcher tool to simulate authenticated registered users and establish user identity in the session data.

2. Conversational Core & Memory

- **User Input & Intent Identification:** Implement the natural language processing to handle user text input and classify queries into the Account Opening intent.
- **Memory:** Implement both the short-term sliding context window for the active chat and the long-term persistent history store for past conversations.
- **Conversation State Management:** Build the basic state management logic to handle standard multi-turn interactions for the Account Opening workflow.

3. Response Generation & Safety

- **Response Generation:** Develop the generation capabilities to guide users interactively, ensuring the bot strictly provides allowlisted deep-links and reduce hallucination by prompt.
- **Deterministic Gate:** Implement the first half of the safety pipeline. This rule-based gate must successfully check for sensitive words, verify the deep-link allowlist, and perform exact number substitution into the chat.

Challenges Faced:

Hardware constraint for MVP: The MVP is originally planned to be running on a MacBook of 16G Memory. It requires the Mac to provide simultaneously the following services: webserver, state management, vector database, structural database, LLM of 2 models and FAQ knowledge store. This puts too much pressure on the local hardware, especially the memory, it will not only influence response latency but also force me to downgrade to a simpler model which will directly influence the performance of metrics evaluated on the project.

Risks involved in the project: This is a bank-industry standard chatbot that is strictly regulated by MAS and other regulations to ensure the safety of bank customers. In spite of achieving the conversational capabilities and smart intent identification and solution, the incremental risks introduced by the LLM should also be mitigated. This includes personal data protection, transaction protection, accuracy in response and adherence to MAS compliance regulations. **It is very hard to achieve absolute compliance semantically. This is the core challenge.**

Plans to overcome challenges:

Hardware constraint for MVP: To offload the workload from the local server, I move the two LLM models to remote by using Groq API services to interact with the model. This is to make the MVP feasible in terms of hardware only. It is not recommended to use a remote LLM for the industry implementation for compliance and safety concerns.

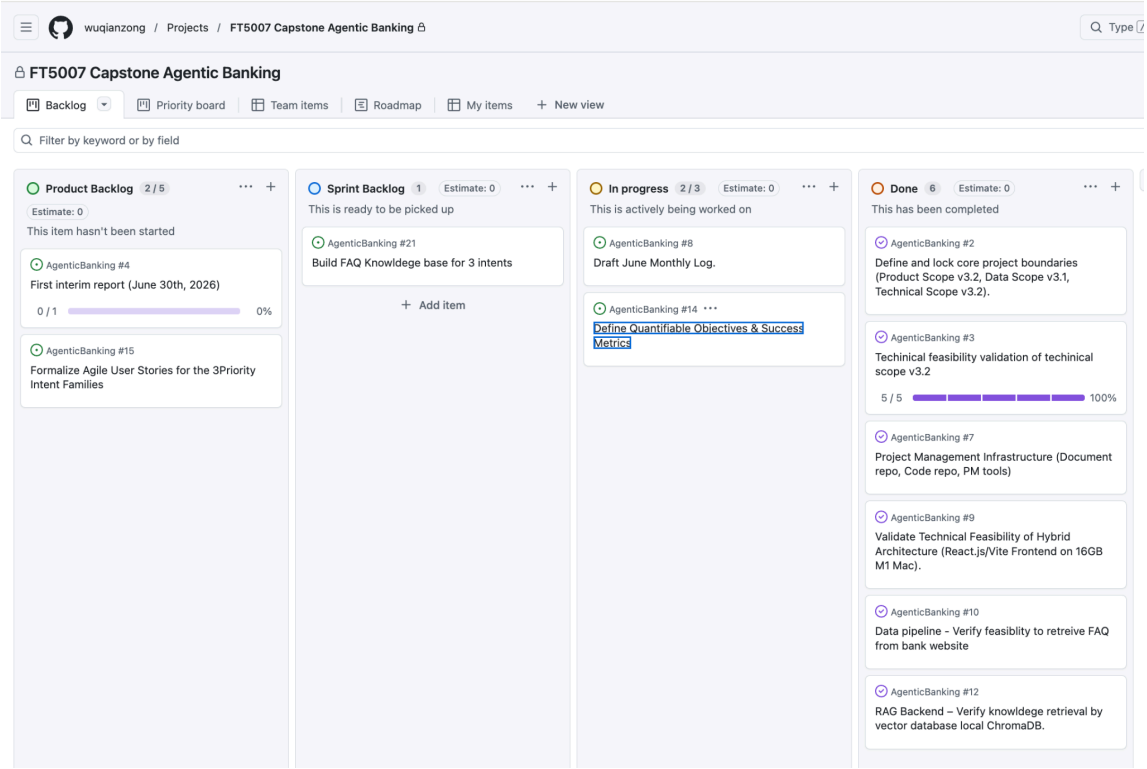
Risks involved in the project: To mitigate the incremental risks, I designed two safety gates, deterministic gate and semantic gate. The deterministic gate to check allow-listed transactional links, filter sensitive words and substitute important number placeholders for accuracy. The semantic gate which is the most challenging part of this project, is to verify the compliance and relevance of the response semantically. Because the semantic gate works on probability, it is very hard to ensure 100% compliance and relevance. My solutions are:

- **Independent judge:** to apply another model for verification, that is to day, use model A to generate response and use model B to verify the response. I hope this can prevent some model bias in the verification process.
- **Verify intent and response:** Another solution is to verify not only the response but also the input of the user to detect any manipulative or malicious intents. But it is largely dependent on the verification prompt and the complexity of the model we choose. The real result is yet to be found in the evaluation stage. This could lead to a dedicated compliance LLM model training.
- **Full chain verification (out of scope):** The third solution is to verify not only the current but also the chain of conversation. Sometimes one response is compliant but if you put the whole conversation chain into consideration, it may be slightly suggesting investment advice or something none-compliant. So it's important to verify the chain instead of the individual response. Due to the hardware restraint. This is out of scope.

Appendix:

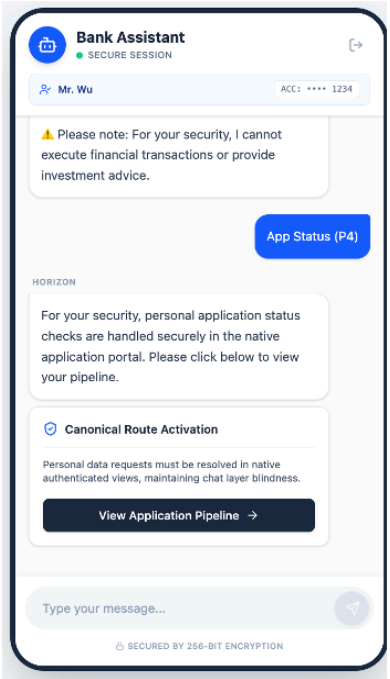
Project Backlog:

https://github.com/users/wuqianzong/projects/2/views/1

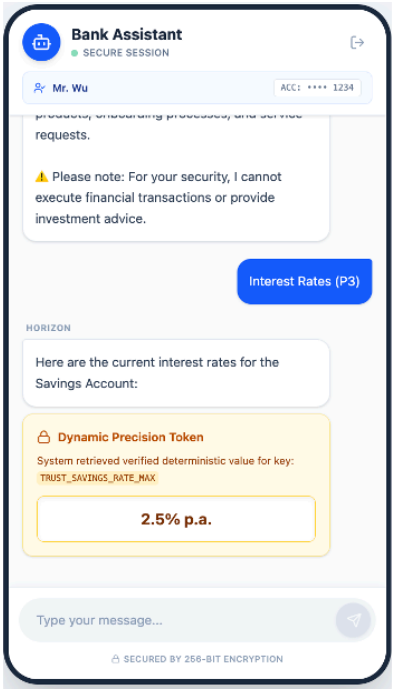


UI of the Proposed MVP:
Note that this is not the final UI. It's for Academic Advisor to have a general idea of the UI.

Allow-listed transaction link:



Precision substitution:



Human Escalation:

